

Formative R activities: A Small COVID-19 Dataset

1. Objective

This project is a series of exploratory data analysis and statistical activities conducted in R using the *COVID-19 India Cases (January–March 2020)* dataset sourced from Kaggle (Teloo, A. (2021)). The dataset contains early COVID-19 reporting data across Indian states and foreign territories during the initial phase of the pandemic, including confirmed cases among Indian and foreign nationals, recovered cases, and deaths.

The activities focused on developing practical skills in data exploration, descriptive statistics, creating categorical variables, frequency analysis, and data visualisation, followed by inferential statistics and probability calculations. Data analysis was performed using R.

These exercises supported the learning outcomes of developing a systematic understanding of key mathematical and statistical concepts underpinning Data Science and Artificial Intelligence, and applying statistical methods to support analytical reasoning and data-driven decision-making. The work also contributed to developing practical experience in reproducible analytical workflows, interpreting statistical outputs, and visualising data insights.

2. Dataset Overview and Initial Exploration

This section includes the number of observations and variables, variable names and types, unique states/territories, the date range, and the most frequently reported state.

2.1. Dataset Structure

The imported 'covid' dataset has **270 observations and 7 variables**; however, an initial data inspection showed that the **Sno** index was created during the upload and does not add value to this dataset; hence, it was removed, resulting in 6 variables, of which 4 were in numeric format and 2 in character/text format. As the **Date** variable was not formatted as a date but as a character, it had to be formatted to the right date format ("**%Y-%m-%d**"). The final dataset contains 6 variables: 4 numeric, 1 date, and 1 character.

	Sno	Date	State/UnionTerritory	ConfirmedIndianNational	ConfirmedForeignNational	Cured	Deaths
1	1	30-01-2020	Kerala	1	0	0	0
2	2	31-01-2020	Kerala	1	0	0	0
3	3	01-02-2020	Kerala	2	0	0	0
4	4	02-02-2020	Kerala	3	0	0	0
5	5	03-02-2020	Kerala	3	0	0	0
6	6	04-02-2020	Kerala	3	0	0	0
7	7	05-02-2020	Kerala	3	0	0	0
8	8	06-02-2020	Kerala	3	0	0	0
9	9	07-02-2020	Kerala	3	0	0	0
10	10	08-02-2020	Kerala	3	0	0	0
11	11	09-02-2020	Kerala	3	0	0	0
12	12	10-02-2020	Kerala	3	0	0	0
13	13	11-02-2020	Kerala	3	0	0	0

Table 1. Original Covid dataset

```
> str(covid)
'data.frame': 270 obs. of 7 variables:
 $ Sno      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Date     : chr  "30-01-2020" "31-01-2020" "01-02-2020" "02-02-2020" ...
 $ State.UnionTerritory : chr  "Kerala" "Kerala" "Kerala" "Kerala" ...
 $ ConfirmedIndianNational : int  1 1 2 3 3 3 3 3 3 3 ...
 $ ConfirmedForeignNational: int  0 0 0 0 0 0 0 0 0 0 ...
 $ Cured    : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Deaths  : int  0 0 0 0 0 0 0 0 0 0 ...
```

	Date	State.UnionTerritory	ConfirmedIndianNational	ConfirmedForeignNational	Cured	Deaths
127	2020-03-13	Telengana	1	0	0	0
128	2020-03-13	Uttar Pradesh	10	1	0	0
129	2020-03-13	Union Territory of Ladakh	3	0	0	0
130	2020-03-13	Tamil Nadu	1	0	0	0
131	2020-03-13	Union Territory of Jammu and Kashmir	1	0	0	0
132	2020-03-13	Punjab	1	0	0	0
133	2020-03-13	Karnataka	6	0	0	1
134	2020-03-13	Maharashtra	14	0	0	0
135	2020-03-13	Andhra Pradesh	1	0	0	0
136	2020-03-14	Delhi	7	0	1	1
137	2020-03-14	Haryana	0	14	0	0
138	2020-03-14	Kerala	19	0	3	0
139	2020-03-14	Rajasthan	1	2	1	0
140	2020-03-14	Telengana	1	0	0	0
141	2020-03-14	Uttar Pradesh	11	1	5	0

Table 2. Cleaned Covid dataset

```

> # 10. Data format final version
> str(covid)
'data.frame': 270 obs. of 6 variables:
 $ Date      : Date, format: "2020-01-30" "2020-01-31" "2020-02-01" "2020-02-02" ...
 $ State.UnionTerritory : chr  "Kerala" "Kerala" "Kerala" "Kerala" ...
 $ ConfirmedIndianNational : int  1 1 2 3 3 3 3 3 3 ...
 $ ConfirmedForeignNational: int  0 0 0 0 0 0 0 0 0 ...
 $ Cured      : int  0 0 0 0 0 0 0 0 0 ...
 $ Deaths    : int  0 0 0 0 0 0 0 0 0 ...
> summary(covid)
      Date      State.UnionTerritory ConfirmedIndianNational ConfirmedForeignNational      Cured
Min.   :2020-01-30 Length   :270 Min.   : 0.000 Min.   : 0.00 Min.   :0.00000
1st Qu.:2020-03-08 N.unique : 27 1st Qu.: 1.000 1st Qu.: 0.00 1st Qu.:0.00000
Median :2020-03-13 N.blank  : 0 Median : 3.000 Median : 0.00 Median :0.00000
Mean   :2020-03-10 Min.nchar: 5 Mean   : 5.552 Mean   : 1.33 Mean   :0.6037
3rd Qu.:2020-03-18 Max.nchar: 36 3rd Qu.: 6.000 3rd Qu.: 0.00 3rd Qu.:0.00000
Max.   :2020-03-21 Max.   :60.000 Max.   :14.00 Max.   :9.00000
Deaths
Min.   :0.00000
1st Qu.:0.00000
Median :0.00000
Mean   :0.09259
3rd Qu.:0.00000
Max.   :1.00000

```

2.2. Data Characteristics

- The COVID data set contains **27 unique states/union territories**
- The data range covered in the data is from **2020-01-30 until 2020-03-21**
- The most frequent cases were represented by **Kerala with 52**

```

> # 8. Date range covered
> range(covid$Date)
[1] "2020-01-30" "2020-03-21"
> min(covid$Date)
[1] "2020-01-30"
> max(covid$Date)
[1] "2020-03-21"
> # -----
>
> # 9. Number of unique states/union territories
> length(unique(covid$State.UnionTerritory))
[1] 27
>
> # 10. Most frequent state
> state_freq <- table(covid$State.UnionTerritory)
> sort(state_freq, decreasing = TRUE)[1]
Kerala
52
..

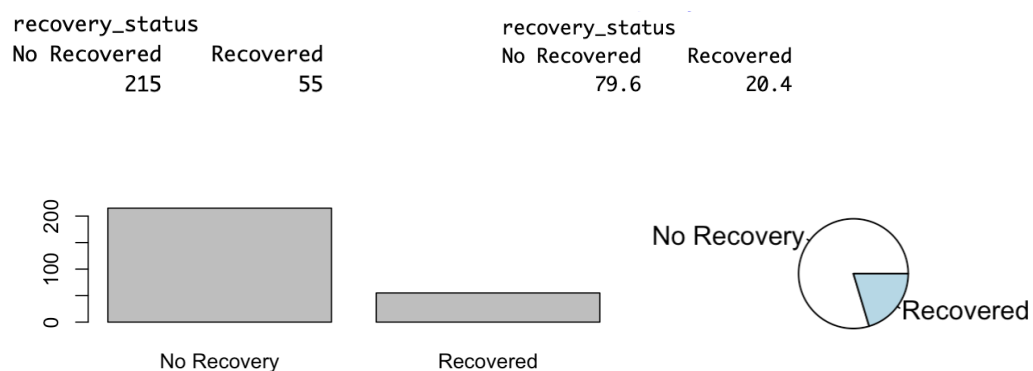
```

3. Frequency Analysis, Binary Variable and Factor Variable

This section provides frequency tables for categorical analysis and reports on recoveries and deaths by creating the `has_recovery` and `has_deaths` variables, and includes bar plots and pie charts.

3.1. Recovered cases

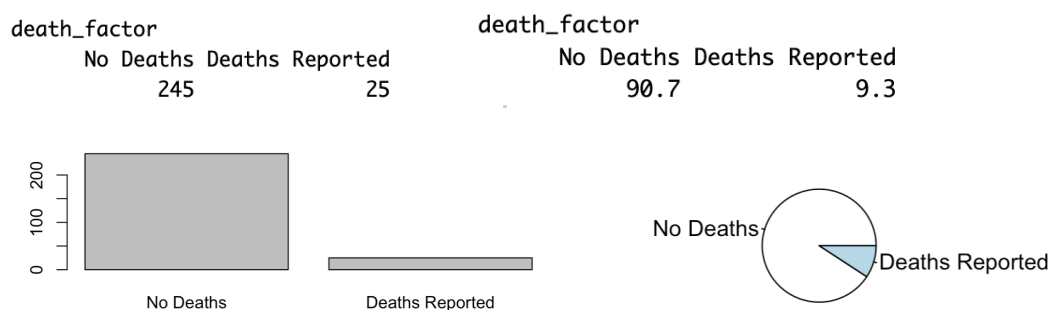
The frequency table for the variable `has_recovery`, which was transformed into a labelled categorical variable for easier interpretation, shows 215 non-recovered vs 55 Recovered cases, with 80% non-recovered.



Figures 1. Bar and plot charts for recovered cases

3.2. Death cases

Frequency tables for the `has_deaths` variable, transformed into a factor, show 245 non-death cases and 25 death cases, with 90% non-death cases being non-death.



Figures 2. Bar and plot charts for death cases

4. Descriptive Statistics and Measures of Dispersion

This section includes **information on Central tendency** (mean, median, mode) and **Dispersion** (variance, standard deviation, and range) **across Indian states** ([ConfirmedIndianNational](#)), along with corresponding visualisations: histograms with density curves for each variable, boxplots to identify outliers, and violin plots to show the shape of the distribution.

4.1. Central tendency and Dispersion for [ConfirmedIndianNational](#)

- **Central tendency:** “*What is typical?*”
 - **Mean:** average number of confirmed cases: **5.6**
 - **Median:** middle value of cases: **3**
 - **Mode:** most common case count: **1 occurs most often**
- **Dispersion:** “*How spread out is the data?*”
 - **Range:** **60** (min to max cases: **0-60**)
 - **Variance:** measures how close the numbers are to the mean (*defined as the average squared difference of each number from the mean*): **69.5**
 - **Standard deviation** (sd): measures how much individual numbers differ on average, from the mean (*defined as the square root of variance*): **8.3**

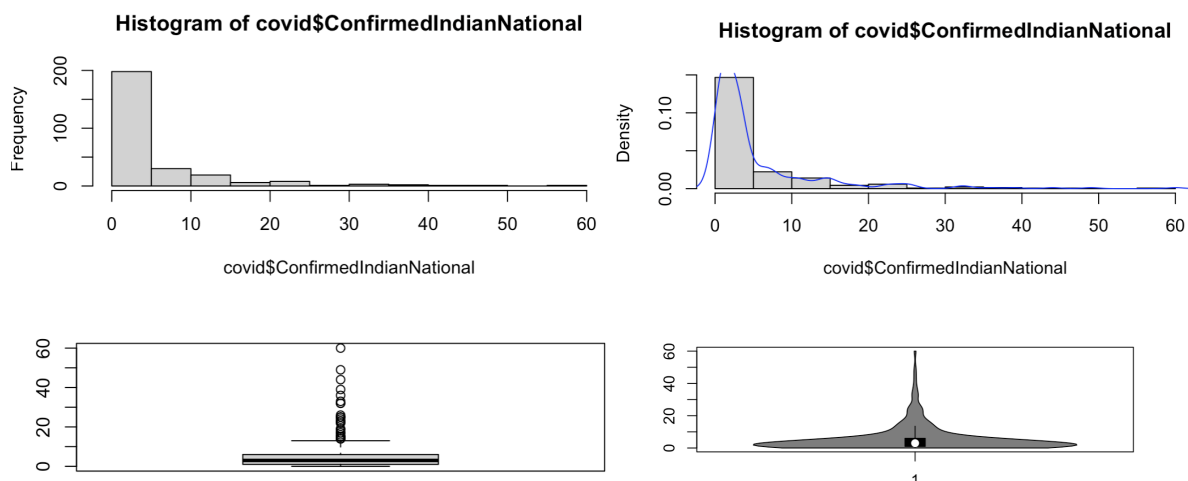


Figure 3. Histograms with density curves, boxplots and violin plots for [ConfirmedIndianNational](#) cases

The measures of central tendency indicate that the typical number of confirmed Indian national Covid cases per observation was relatively low during the analysed period. While the average number of confirmed cases was 5.6, the median value of 3 and the mode of 1 suggest that most observations contained only a small number of cases. The mean is notably higher than both the median and the mode, indicating a positively skewed distribution because a small number of observations with high case counts raise the overall average.

The measures of dispersion show substantial variability in the dataset. The large variance of 69.5 and the standard deviation of 8.3 were higher than the mean of 5.6, indicating that case numbers varied considerably across observations. The wide range of 0 to 60 further confirms the presence of both very low and relatively high case counts within the dataset. Overall, although most observations reported low case counts, a smaller number reported significantly higher counts, contributing to a positively (right)- skewed distribution with notable outliers in the visualisations.

4.2. Visualisation for other variables

The visualisations across the four variables (**ConfirmedIndianNational**, **ConfirmedForeignNational**, **Cured**, and **Deaths**) showed that all distributions were positively skewed, with most observations concentrated at low values and a smaller number of observations containing much higher counts. The boxplots revealed multiple outliers, particularly for confirmed cases and recoveries, indicating variability across observations and states. Histograms and violin plots further highlighted that deaths were heavily concentrated near zero, while confirmed cases displayed the greatest spread and variability. Overall, the visualisations demonstrated that the early Covid data were highly unevenly distributed, with a few observations contributing disproportionately higher case counts.

4.2.1. ConfirmedForeignNational

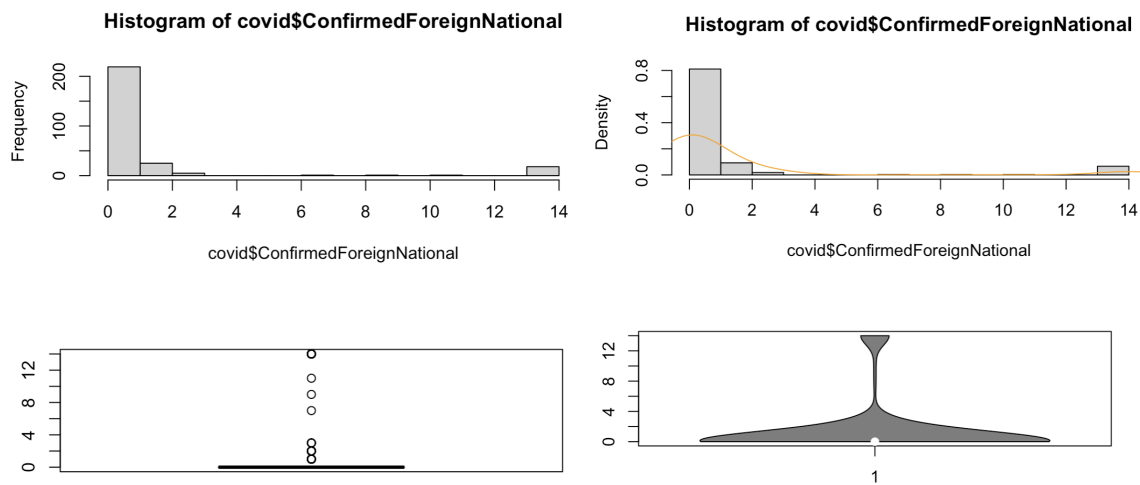


Figure 4. Histograms with density curves, boxplots and violin plots for ConfirmedForeignNational cases

4.2.2. Cured

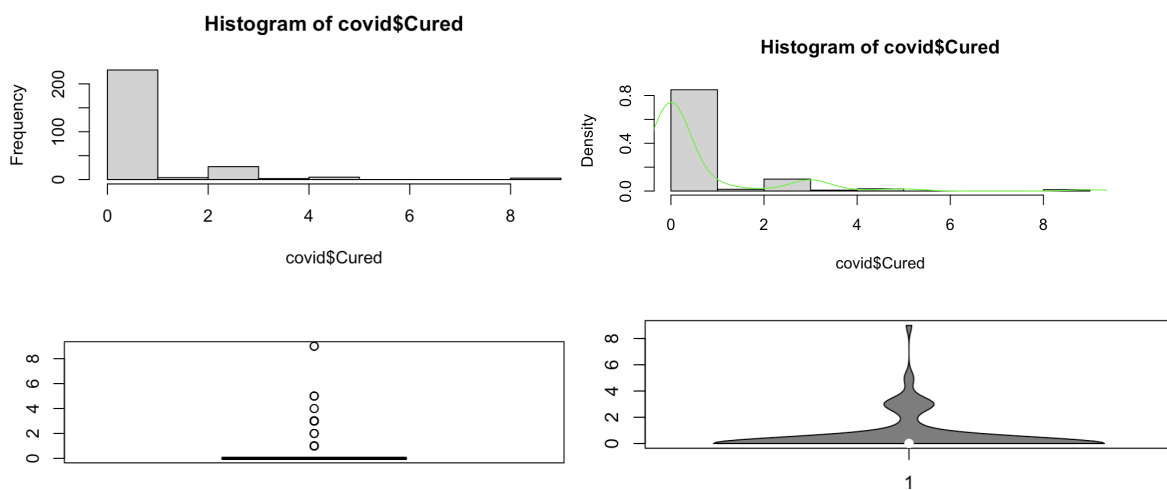


Figure 5. Histograms with density curves, boxplots and violin plots for Cured cases

4.2.3. Deaths

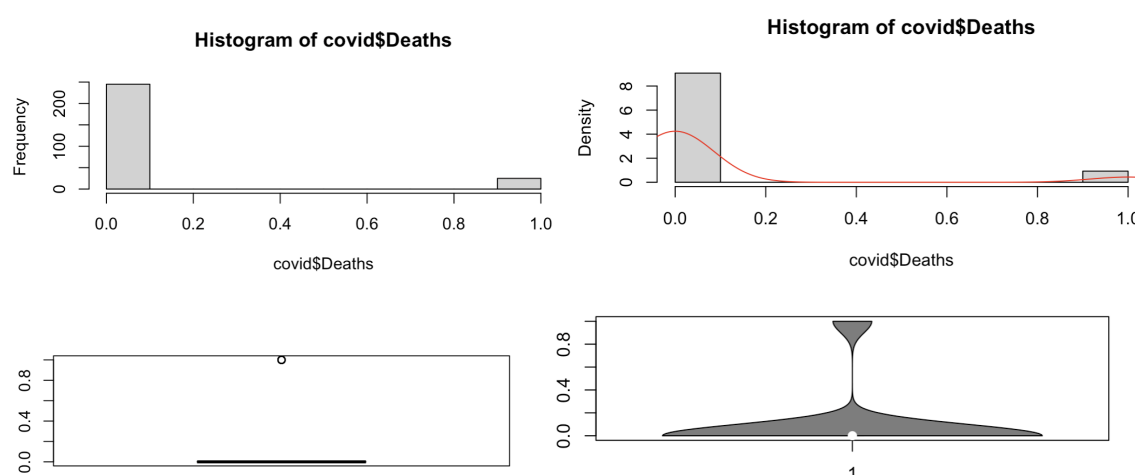


Figure 6. Histograms with density curves, boxplots and violin plots for Deaths cases

5. Categorical Factor Variables and Contingency Table

This section reports on the total_cases variable, case_level categories, factor levels and frequency table for the State/Union Territory variable and identifies the top 10 states with the highest number of daily reports.

5.1. Total cases and case severity variables

Total cases variable combined Indian and Foreign nationals data, and **case severity categorical variables** (No Cases, Low Cases, Medium Cases, High Cases) were defined based on the predefined criteria, resulting in 177 “low cases”, 61 “medium cases”, 32 “high cases” and 0 “no cases” reported.

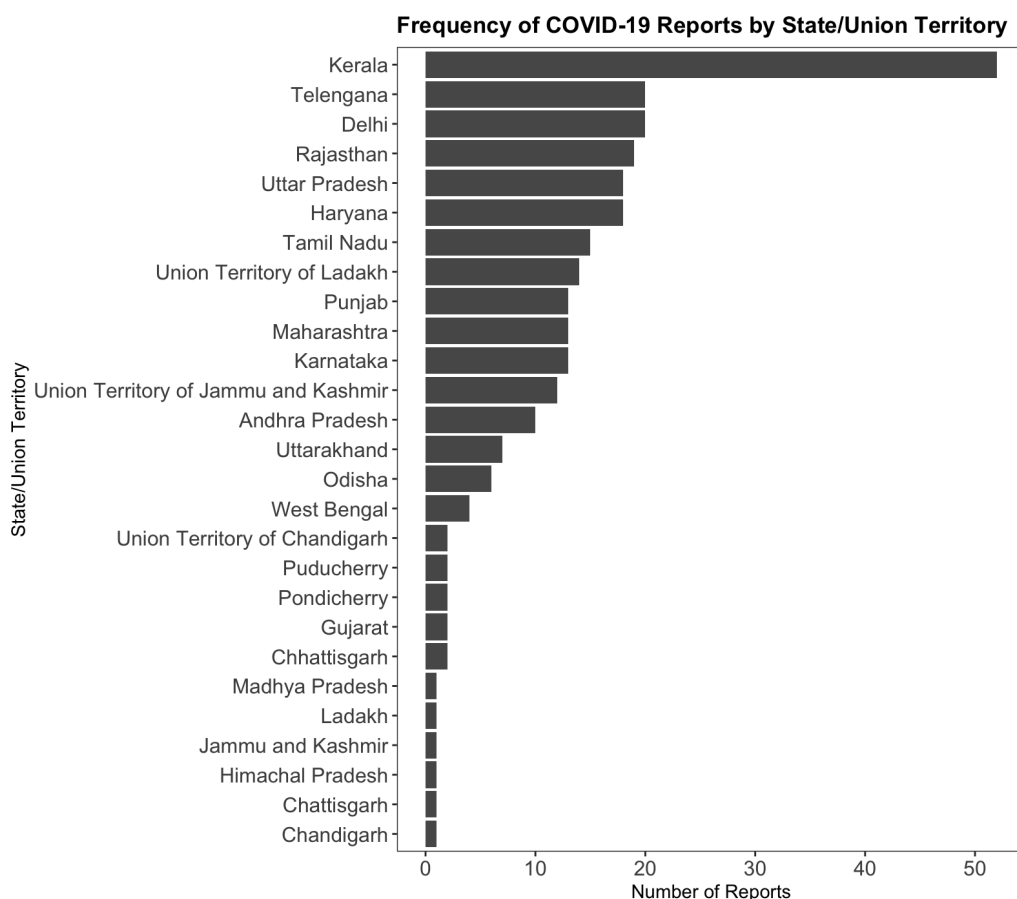
Predefined criteria	Result
No Cases = 0	0
Low Cases = 1-5 cases	177
Medium Cases = 6-15 cases	61
High Cases - 16+ cases	32

Table 1. Covid severity results

5.2. Top 10 states with the highest number of daily reports, including frequency of reported cases by State/Union Territory

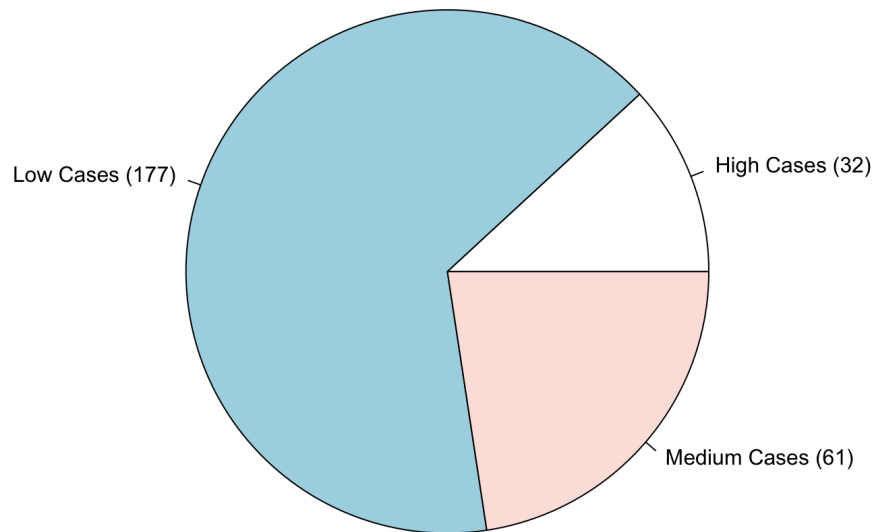
Kerala led in reported cases, with 52 (11 high, 3 medium, and 38 low), followed by Karnataka and Maharashtra, each with 13, rounding out the list.

	case_level				Sum
	No Cases	Low Cases	Medium Cases	High Cases	
Kerala	0	38	3	11	52
Delhi	0	10	8	2	20
Telangana	0	16	2	2	20
Rajasthan	0	9	8	2	19
Haryana	0	7	7	4	18
Uttar Pradesh	0	0	14	4	18
Tamil Nadu	0	15	0	0	15
Union Territory of Ladakh	0	10	4	0	14
Karnataka	0	4	9	0	13
Maharashtra	0	3	3	7	13
Sum	0	112	58	32	202



6. Pie chart: Distribution of case severity levels based on total confirmed cases

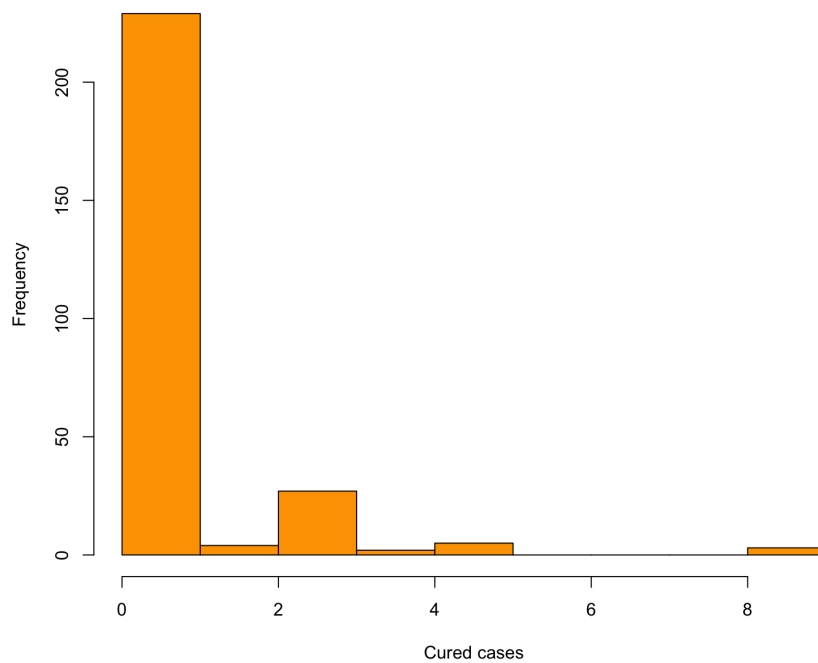
Pie Chart of Distribution of COVID-19 Case Severity Levels



7. Histogram with the distribution of recovery numbers

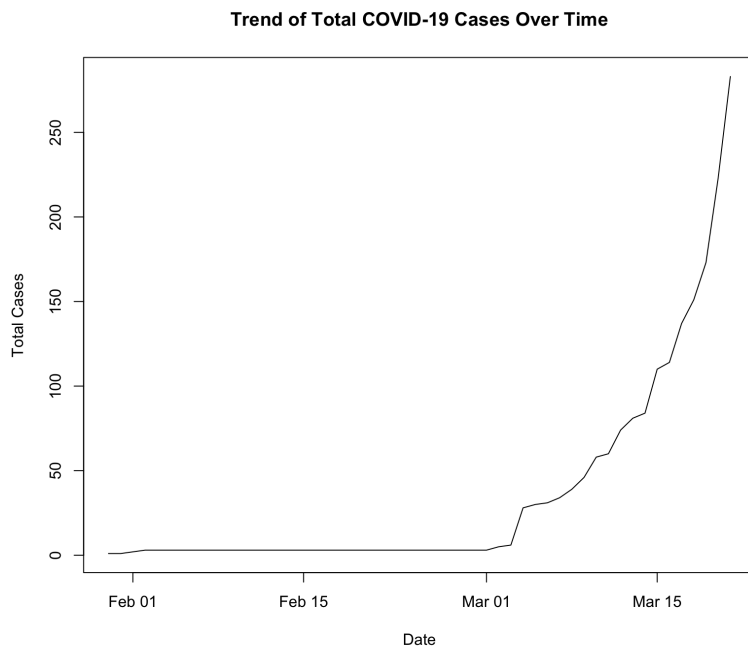
The histogram indicates that the majority of recovery cases are reported as 'zero', with only a few high counts.

Distribution of Recovery Numbers of COVID-19



8. Line chart showing the trend of total cases over time

The surge of total cases is visible in the spike as of early March.



9. Relationship between Region and Case Severity Level (Cross tabulation)

- Kerala had the highest count of High Cases: 11
- Uttar Pradesh has the highest count of Medium Cases: 14
- Kerala had the highest count of Low Cases: 11, followed by Telangana (16) and Tamil Nadu (15)

State.UnionTerritory	case_level					Sum
	No Cases	Low Cases	Medium Cases	High Cases		
Andhra Pradesh	0	10	0	0	0	10
Chandigarh	0	1	0	0	0	1
Chattisgarh	0	1	0	0	0	1
Chhattisgarh	0	2	0	0	0	2
Delhi	0	10	8	2	20	
Gujarat	0	1	1	0	2	
Haryana	0	7	7	4	18	
Himachal Pradesh	0	1	0	0	1	
Jammu and Kashmir	0	1	0	0	1	
Karnataka	0	4	9	0	13	
Kerala	0	38	3	11	52	
Ladakh	0	0	1	0	1	
Madhya Pradesh	0	1	0	0	1	
Maharashtra	0	3	3	7	13	
Odisha	0	6	0	0	6	
Pondicherry	0	2	0	0	2	
Puducherry	0	2	0	0	2	
Punjab	0	12	1	0	13	
Rajasthan	0	9	8	2	19	
Tamil Nadu	0	15	0	0	15	
Telangana	0	16	2	2	20	
Union Territory of Chandigarh	0	2	0	0	2	
Union Territory of Jammu and Kashmir	0	12	0	0	12	
Union Territory of Ladakh	0	10	4	0	14	
Uttar Pradesh	0	0	14	4	18	
Uttarakhand	0	7	0	0	7	
West Bengal	0	4	0	0	4	
Sum	0	177	61	32	270	

10. Relationship between Case Severity Level and Recovery status (Cross tabulation)

Recovery Status differs significantly by Severity, with 88% recovery rate among High Cases, 62% among Medium Cases, and 28% among Low Cases.

srecovery_status	case_level					
	Recovered	No Recovered	Sum	Recovered	No Recovered	
No Cases	0	0	0	No Cases		
Low Cases	16	161	177	Low Cases	28.0	72.0
Medium Cases	18	43	61	Medium Cases	62.1	37.9
High Cases	21	11	32	High Cases	88.2	11.8
Sum	55	215	270			

Absolute recovery cases Recovery Rate

11. Bayes probability calculations

Out of 270 daily state reports:	Reports with recoveries: 55 total reports showed patient recoveries
Kerala reports: 52 reports Delhi reports: 20 reports	Kerala reports with recoveries: 19 reports Delhi reports with recoveries: 8 reports

K - report came from Kerala

D - report came from Delhi

R - report with recoveries

The probability that a randomly selected report comes from Kerala: $P(\text{Kerala})$

$$P(K) = 52 / 270 = 0.193 \rightarrow 19.3\%$$

The probability that a randomly selected report shows recoveries: $P(\text{Recovery})$

$$P(R) = 55 / 270 = 0.204 \rightarrow 20.4\%$$

The probability that a Kerala report shows recoveries: $P(\text{Recovery} | \text{Kerala})$

$$P(R | K) = 19 / 52 = 0.365 \rightarrow 36.5\%$$

The probability that a Delhi report shows recoveries: $P(\text{Recovery} | \text{Delhi})$

$$P(R | D) = 8 / 20 = 0.40 \rightarrow 40\%$$

If a report shows patient recoveries, what is the probability that it came from Kerala? $P(\text{Kerala} | \text{Recovery})$

$$P(\text{Kerala} | \text{Recovery}) = P(R | K) * P(K) / P(R)$$

$$0.365 * 0.193 / 0.204 = 0.345 \rightarrow 34.5\%$$

$$19 / 55 = \rightarrow 34.5\%$$

Using Bayes' theorem, the probability that a report came from Kerala given that it showed patient recoveries was 34.55%.

The difference between frequentist and Bayesian probability calculations

Frequentist approach Relying on the observed data	Bayesian approach Updating the information based on new evidence
What's the probability of recovery if I know the report is from Kerala? Conditional probability = ("Recovery given Kerala")	What's the probability the report is from Kerala if I know it shows recovery? Bayes' theorem = "Kerala given Recovery" Reverse question

12. Key Learnings and Reflection

- Through these activities, I have developed a more structured and reproducible workflow in R by moving from ad hoc console experimentation to more organised, script-based analysis. I have learned the importance of maintaining clean, reusable scripts, separating raw and cleaned datasets, and organising project folders to support reproducibility and facilitate debugging. Additionally, clear variables/object names supported debugging, i.e., (tb, df)
- The exercises also highlighted the importance of data cleaning before analysis, particularly when handling issues such as incorrect date formats, unnecessary index variables, and inconsistent variable types.
- In addition, I gained practical experience working with summary tables, factors, categorical variables, and frequency and contingency tables to analyse reporting patterns across states and case categories, and to sort and summarise grouped data.
- Summary tables: reading summary output describing variable distribution and deciding how it gets meaningfully is another challenge. Specifically deciding whether continuous variables should be reported using the mean and standard deviations, or median and interquartile range, depending on their individual distribution, and whether categorical variables should be presented by the counts, percentages, or both, suited and ordered. I learned that the choice of summary table should be based on the variable type and distribution, and aligned with the purpose of the downstream analysis.
- Contingency tables describe relationships between 2 categorical variables. The order of the variables matters for interpretation and must be guided by the research question. Placing the grouping variable in the rows and the outcome variable in the columns makes row percentages easier to interpret and helps ensure the analysis answers the intended question clearly. The guiding principle is that the explanatory (grouping) variable comes first, and the outcome variable second.

- By calculating descriptive statistics and creating visualisations such as histograms, boxplots, and violin plots, I improved my understanding of measures of central tendency, dispersion, skewness, and outliers. Moreover, data trends are presented using a line chart and the FUN function to aggregate data by date, and distributions are shown with the ggplot2 library, pie charts, and histograms.
- Frequentist vs Bayesian probabilities: While basic probabilities describe the observed frequencies in the dataset, Bayes' theorem addresses a different question by inverting the conditional probability to estimate the likelihood of an event after considering new evidence.

Overall, the activities helped me move from having no R knowledge and basic mathematical and statistical analysis to a more systematic, analytical approach to data exploration and interpretation.

Reference

Teloo, A. (2021) *A Small COVID-19 Dataset*. Kaggle. Available at: <https://www.kaggle.com/datasets/aditeloo/a-small-covid19-dataset> (Accessed: 15 May 2026)

Bibliography

Bader, M. and Leuzinger, S. (2025) *R-ticulate: A Beginner's Guide to Data Analysis for Natural Scientists*. 1st edn. Hoboken: Wiley.